

Tamoxifen and Flaxseed Alter Angiogenesis Regulators in Normal Human Breast Tissue In Vivo

The incidence of breast cancer is increasing in the Western world and there is an urgent need for studies of the mechanisms of sex steroids in order to develop novel preventive strategies. Diet modifications may be among the means for breast cancer prevention. Angiogenesis, key in tumor progression, is regulated by the balance between pro- and anti-angiogenic factors, which are controlled in the extracellular space. Sampling of these molecules at their bioactive compartment is therefore needed. The aims of this study were to explore if tamoxifen, one of the most used anti-estrogen treatments for breast cancer affected some of the most important endogenous angiogenesis regulators, vascular endothelial growth factor (VEGF), angiogenin, and endostatin in normal breast tissue in vivo and if a diet supplementation with flaxseed had similar effects as tamoxifen in the breast. Microdialysis was used for in situ sampling of extracellular proteins in normal breast tissue of women before and after six weeks of tamoxifen treatment or before and after addition of 25 g/day of ground flaxseed to the diet or in control women. We show significant correlations between estradiol and levels of VEGF, angiogenin, and endostatin in vivo, which was verified in ex vivo breast tissue culture. Moreover, tamoxifen decreased the levels of VEGF and angiogenin in the breast whereas endostatin increased significantly. Flaxseed did not alter VEGF or angiogenin levels but similar to tamoxifen the levels of endostatin increased significantly. We conclude that one of the mechanisms of tamoxifen in normal breast tissue include tipping of the angiogenic balance into an anti-angiogenic state and that flaxseed has limited effects on the pro-angiogenic factors whereas the anti-angiogenic endostatin may be modified by diet. Further studies of diet modifications for breast cancer prevention are warranted.